

JEE MAIN REVISION MASTER: CIRCLE SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 183 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Circle - Core Application	183 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circle - Core Application (Very High Yield — 183 PYQs)

Core Principle: Key problem pattern in Circle.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let C be a circle having centre in the first quadrant and touching the x -axis at a distance of 3 units from the origin. If the circle C ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x+(k-1)y+3=0$ and $2x+...$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the set of all values of r , for which the circles $(x+1)^2+(y+4)^2=r^2$ and $x^2+y^2-4x-2y-4=0$ intersect at two distinct p ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x+y=1$ is $\sqrt{14}$, then the square of the radius of C is _____.

Correct Answer: 8

Solution Analysis: $r^2 = (\frac{\sqrt{14}}{2})^2 + (r - \frac{1}{\sqrt{2}})^2$ $r^2 = (\frac{7}{2}) + r^2 - \sqrt{2}r + (\frac{1}{2})$ $\sqrt{2}r = 4$ $r = 2\sqrt{2}$ $r^2 = 8$